Project Reprort

Optimizing Revenue Leakage in the Hospitality Sector.

A Data-Driven Consulting Project

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Project created using Python and Tableau

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1.Objective

Atliq Hospitality, a multi-city hotel chain, observed declining profits across several properties. Despite having access to detailed booking and service data, the group lacked actionable insights to identify revenue leakages and underutilized services.

This project aimed to:

- Detect and quantify revenue leakage across hotels
- Understand booking behavior by time, platform, and room type
- Use KPI analysis and statistical testing to validate key trends
- Identify underperforming hotels and group them into strategy segments
- Deliver a consulting-style recommendation plan and a Tableau dashboard for ongoing decision-making

2.Data Exploration

The dataset was structured around a star schema with five core tables:

Table Name	Description
fact_bookings	Daily booking-level data per hotel
fact_aggregated_bookings	Hotel-level metrics per day (occupancy, revenue)
dim_date	Calendar-level mapping
dim_rooms	Room capacity and types
dim_hotel	Hotel details — name, city, type, etc.

Key Preprocessing Steps:

- Combined property_name + city in dim_hotel to create a unique hotel identifier (since same hotel name existed across multiple cities)
- Identified primary keys across tables
- Changed date columns to datetime objects
- Ensured no data integrity issues (e.g., bookings never exceeded room capacity)
- Created derived columns for analysis:
  - occupancy_percentage = bookings / rooms
  - revenue_leakage = revenue_generated – revenue_leakage
  - has_rated = 1 if rating available, else 0
  - Extracted check_in_day, check_in_month

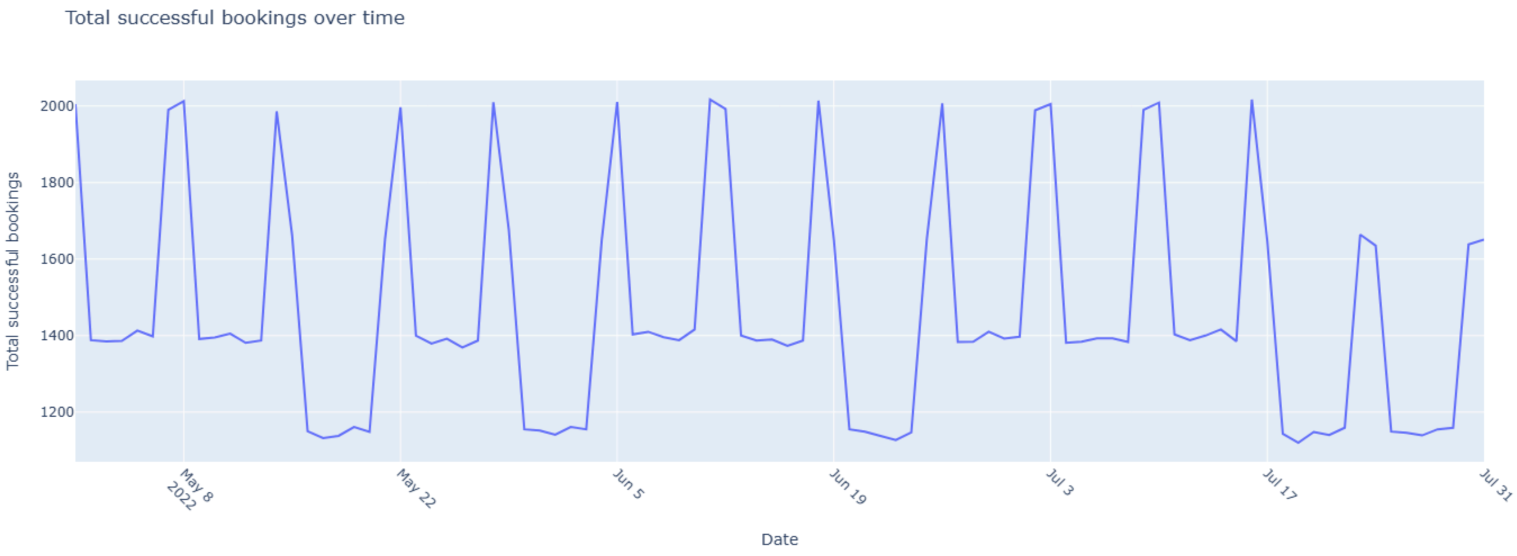
Observations:

- Data spanned 92 days (May 1 – July 31, 2022)
- 25 hotels across multiple cities, 7 booking platforms
- ~58% of ratings_given were missing, requiring flag-based handling
- 4 room types: Standard, Elite, Premium, Presidential
- Revenue leakage happens only when booking is cancelled(No refund is issued if status is 'No Show')

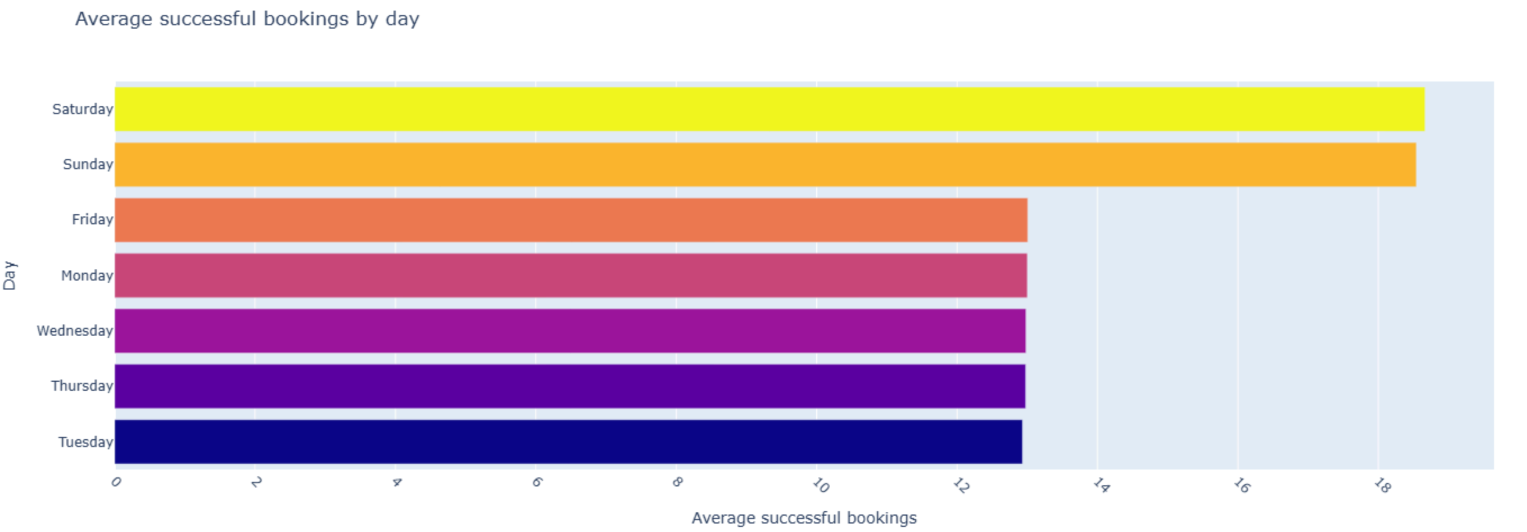
3.Exploratory Data Analysis

Bookings Trend:

The data encompasses bookings across various hotels from May 1 to July 31, 2022.



We see a sharp incline in bookings on certain days and a sudden fall indicating increased bookings on weekends



This chart shows weekend bookings exceeded weekday bookings.

To confirm this we perform Mann-Whitney U test.

Null hypothesis: Bookings on weekdays and weekends come from the same distributions

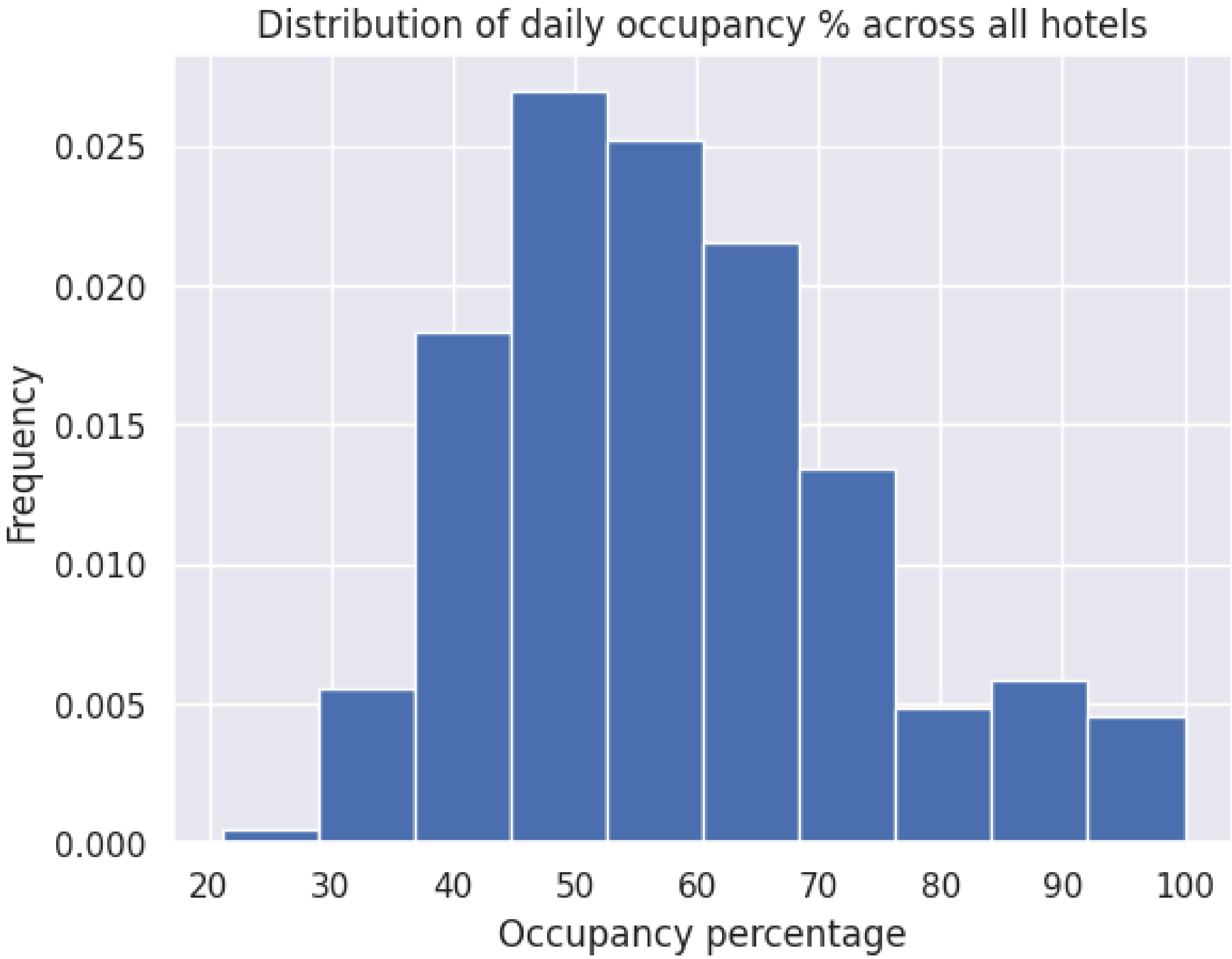
Alternative hypothesis: Weekend bookings are generally higher.

stat: 12109530.0

p: 4.881e-182

The above test confirms that distribution of bookings on weekends are significantly higher than bookings on weekdays.

Occupancy Insights:

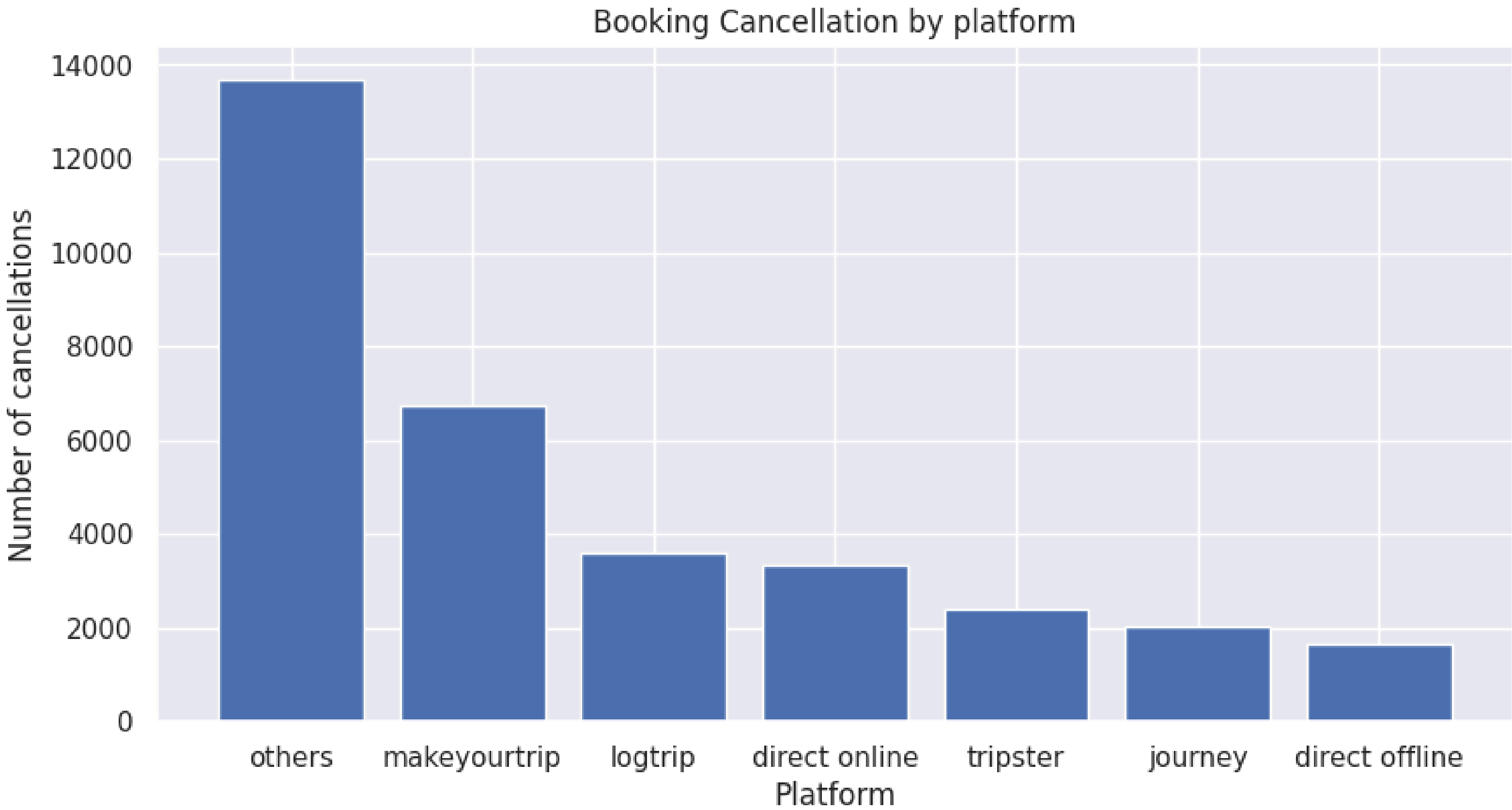


Across all hotels the daily occupancy percentage are expected to be around 45-65%.

Top 5 Hotels		Bottom 5 Hotels	
Property	**Occ%**	**Property**	**Occ%**
Atliq Blu, Mumbai	68.27%	Atliq city, Mumbai	52.93%
Atliq Grands, Delhi	67.69%	Atliq Bay, Mumbai	45.45%
Atliq City, Hyderabad	66.37%	Atliq Grands, Banglore	44.48%
Atliq Palace, Delhi	66.35%	Atliq Exotica, Heyderabad	44.71%
Atliq Exotica, Mumbai	66.20%	Atliq Seasons, Mumbai	44.50%

Platform performance:

- There are six primary booking channels available across all hotels.
- All other minor booking channels are grouped under ‘others’.
- Approximately 42% of users across all channels have left a rating.



- The “Makeyoutrip” platform accounts for the highest number of cancellations, making it the leading contributor for revenue leakage.

City-wise and hotel type Analysis:

- **Atliq hospitality has hotel chain across four cities : Delhi, Hyderabad, Bangalore and Mumbai.**

  Is there one city among the four that experiences greater revenue leakage than the others?

  To test this we performed Kruskal–Wallis test:

  Null Hypothesis: All groups come from the same distribution (i.e., no difference in medians/ranks)

  Alternative Hypothesis: At least one group is different from the others

  stat: 13.220769230769235

  p: 0.004182707684133614

  Even though the p-value is not considerably lower, we will continue with our post hoc analysis.

  The results indicate that Mumbai experiences higher revenue leakage compared to all other cities.

- **Atliq Hospitality encompasses two types of hotels: Luxury and Business.**

  Is there a difference in revenue leakage between these two categories?

  We perform Mann-Whitney U test:

  Hotel Type vs leakage

  Ho: Nodifference in central tendency/rank of two groups.

  stat: 61.0

  p: 0.5522142220341504

  There is insufficient evidence to reject the null hypothesis.

4.Root Cause Analysis

Initial Approach:

We Started by sorting hotels based on total revenue to find “low performers.”

What Went Wrong:

- The lowest revenue hotel : Atliq Grands Delhi ,actually showed minimal revenue leakage
- It had low room inventory but high occupancy and ADR, meaning it was efficiently priced and utilized
- Conclusion: Low revenue ≠ Poor performance

Final Approach Leakage based diagnosis:

Method:

- Computed Revenue Leakage = (ADR × Rooms) − Revenue Generated
- Aggregated leakage at hotel level
- Ranked hotels to identify top leakage contributors

It turns out that Atliq Grands, Delhi, the hotel generating the lowest revenue, actually experiences the least revenue leakage.

	Least Efficient		Most Efficient
Property name	Atliq Seasons, Mumbai	Atliq City ,Mumbai	Atliq Grands, Delhi
Revenue Realized	66.12M	87.99M	36.06M
ADR	₹16,606.10	₹14,634.23	₹11,437.10
RevPAR	₹7,409.85	₹7,776.27	₹7,537.00
ALOS	1.42	1.44	1.43
Average Rating	2.29	3.04	4.25
Booking % of Standard Rooms	16.25	32.28	36.22
Booking % of Elite Rooms	41.34	36.94	42.06
Booking % of Premium Rooms	25.11	20.47	15.16
Booking % of Presidential Rooms	17.3	10.31	6.57
Revenue % from Standard Rooms	9.52	21.46	25.14
Revenue % from Elite Rooms	33.41	33.5	40.8
Revenue % from Premium Rooms	27.18	25.05	20.1
Revenue % from Presidential Rooms	29.89	19.99	13.97

Cancellation Rate	24.79	25.06	25.06
No Show Rate	4.62	5.39	4.92
Occupancy %	44.51	52.94	67.7

Key findings:

Occupancy Percentage:

The most efficient hotel operated at **67.7% occupancy**, whereas the least efficient hotels reported significantly lower rates — **52.9% and 44.5%**, respectively.
➤ *This indicates that revenue potential was not being fully utilized in the inefficient properties.*

Customer Satisfaction:

The average rating for the most efficient hotel stood at **4.25**, while the inefficient hotels recorded notably lower scores — **3.04 and 2.29**.
➤ *Lower guest satisfaction may contribute to fewer repeat bookings and weaker brand trust.*

Room Type Utilization:

The inefficient hotel 17564 relied heavily on **luxury rooms (RT4)**, which made up **17.3% of bookings but contributed 29.9% of revenue**. In contrast, the efficient hotel had a **more balanced distribution**, with higher engagement in standard and mid-range room types (Standard and Elite).
➤ *The hotel seems to have more high-end rooms, but most guests prefer cheaper options. Since those expensive rooms aren't getting booked, many stay empty , causing revenue loss.*

Cancellations and No-shows:

All three hotels had **similar cancellation (~25%) and no-show rates (~5%)**, ruling out booking drop-off as a primary cause of inefficiency.

Weekday Performance Gaps:

Occupancy in inefficient hotels dropped sharply during **weekdays**, with most bookings clustered around weekends.

➤ ***Lack of weekday demand-generation*** *strategies could be contributing to cumulative leakage*.

City Saturation & Competition:

Both least efficient hotels were **located in Mumbai**, a city with high competition and demand variability.
➤ *Maybe these hotels aren't doing anything special to stand out, or their prices aren't flexible enough. Also, they might be depending too much on the same booking platforms as everyone else*

5.Consulting Recommendation

1. Fix the Room Mix

Some hotels are offering a lot of luxury rooms (Premium and Presidentail), but not enough people are booking them. Instead, more guests seem to prefer standard or mid-range rooms (Standard and Elite).

So, it might help if:

- A few luxury rooms can be changed into more affordable ones
- The number of room types is adjusted based on actual booking trends

This could make sure more rooms get booked and fewer sit empty.

2. Try to Improve Guest Ratings

The hotels with the most leakage also had the lowest guest ratings. Even if rooms are nice, guests might not be happy with the service or overall experience.

Some ideas to improve ratings:

- Ask guests to leave reviews (maybe with a small reward or reminder).
- Improve small things in guest experience like check-in process, cleanliness, or communication

Better ratings could help build trust and bring in more bookings.

3. Reduce Overuse of Certain Platforms

Some hotels depend too much on third-party platforms like “MakeYourTrip”. These platforms often bring in one-time customers and charge commission and make booking cancellation easy contributing in revenue leakage.

Maybe the hotels can:

- Try to get more direct bookings through their own website or offline channels
- Give small discounts or loyalty points to people who book directly
- Use offers on certain platforms only for low-demand days like weekdays

This might help reduce costs and bring more repeat guests.

4. Get More Bookings on Weekdays

A lot of revenue leakage was happening because weekday bookings were too low, especially in business hotels.

These hotels can try things like:

- Making deals with nearby companies for employee stays
- Renting out unused areas as co-working spaces during the day
- Hosting small local events or workshops

This can help fill empty rooms and make better use of hotel space during the week.

5. Focus on Mumbai Hotels

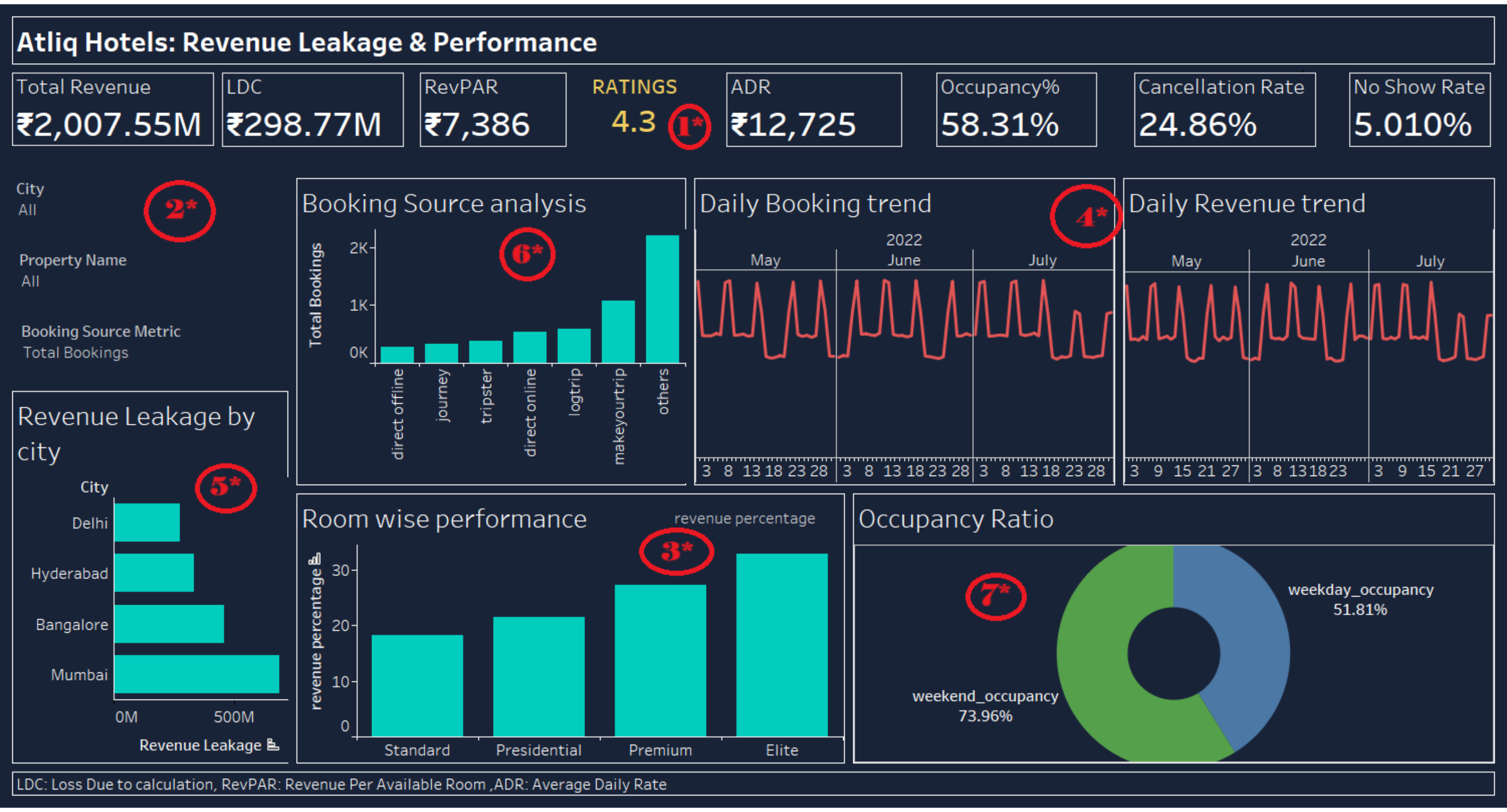
Most hotels with the biggest leakage were in Mumbai. It's a busy market but also very competitive. These hotels might need to do a bit more to stand out.

Some things they could try:

- Adjust prices more often based on demand
- Create packages or promotions for locals, like weekend staycations
- Offer flexible rates for weekdays and long stays

5.Dashboard Overview & User Guide

Dashboard Image:



Elements and Description:

1* - Global KPI card

- Gives an at-a-glance summary of business health.

2* - Filters

- **City Filter** – To view metrics for a specific city
- **Property Name Filter** – To drill down to an individual hotel
- **Booking Source Metric Selector** – Switches between metrics like total bookings or revenue from each booking platform
- Lets the user control which data is shown in the charts and KPIs.

3* - Room-Wise Performance

- Bar chart showing % revenue contribution by room type (Standard, Presidential, Premium, Elite)
- Filter to chose between revenue or booking percentage
- Assesses which room types are more profitable.

4* - Daily Booking and Revenue Trend

- Time series line showing daily number of bookings and revenues from May to July.
- Helps spot weekly cycles, spikes (weekends), and dips (weekdays).
- Tracks revenue patterns and highlights any mismatch with booking patterns.

5* - Revenue Leakage by City

- Compares total revenue leakage across cities
- Identifies cities where leakage is most severe

6* - Booking Source Analysis

- Bar chart showing total bookings by platform (e.g., makeyourtrip, logtrip, tripster, etc.)
- Identifies which booking platforms are used the most and may help uncover platform-specific revenue patterns.
- Can be controlled by Booking Source Metric Selector.

7* - Occupancy Ratio

- Pie chart comparing weekday vs. weekend occupancy.
- Helps understand whether demand is concentrated around weekends.

7.Summary

This project was carried out to help Atliq Hotels identify revenue leakages and improve profitability across its chain of properties. The study involved analyzing booking trends, platform usage, room performance, and guest behavior over a period of 92 days.

Key insights included:

- High revenue leakage was concentrated in Mumbai-based business and luxury hotels
- Over-dependence on low-retention booking platforms and underutilization of luxury inventory contributed to losses
- Weekday occupancy was significantly lower than weekends
- Low-rated hotels had higher leakage and fewer repeat guests

Recommendations were made to rebalance room types, diversify platform usage, target weekday demand, and improve service quality. A dashboard was created to help business users explore trends and KPIs in real-time.

The report concludes with visual documentation, KPI definitions, and suggestions for next steps.

Appendix – KPI Definitions

KPI	Definition
Total Revenue	Sum of all booking revenues per hotel
Revenue Realized	Actual revenue after adjusting for cancellations.
Revenue Leakage	Difference between potential revenue and revenue realized
Occupancy %	(Rooms Occupied / Total Available Rooms) × 100
ADR (Average Daily Rate)	Total Revenue / Rooms Sold
RevPAR	Total Revenue / Rooms Available
ALOS (Average Length of Stay)	Total Nights Stayed / Number of Bookings
Cancellation Rate	(Cancelled Bookings / Total Bookings) × 100
No-Show Rate	(No-Show Bookings / Total Bookings) × 100
Ratings Given	Average rating score left by customers (where available)
Booking %, Revenue % (All rooms)	Share of bookings and revenue from each room type